Software Requirements Specification (SRS)

Employee Management System

1. Introduction

The Employee Management System is a software system designed to manage employee information in an organization. It allows authorized users to add, view, update, and delete employee records. The system also helps in maintaining basic employee details such as employee ID, name, department, designation, contact information, and salary.

2. Problem Statement

Managing employee information manually can be time-consuming and may lead to errors, duplicate records, or difficulty in finding required information. The proposed system aims to provide a simple and centralized solution for storing and managing employee records efficiently.

3. Objectives

- Add and maintain employee information.
- View and search employee records easily.
- Update existing employee details.
- Delete employee records when required.
- Manage basic department and designation information.
- Provide simple and secure access to authorized users.

4. Functional Requirements

- **Employee Registration:** The system shall allow users to add new employees with details such as Employee ID, Name, Department, Designation, Contact, and Salary.
- **View Employees:** The system shall display the list of registered employees.
- **Search Employee:** The system shall allow users to search employees using Employee ID or Name.
- **Update Employee:** The system shall allow authorized users to modify employee information.
- **Delete Employee:** The system shall allow authorized users to remove an employee record.
- **Login:** The system shall provide basic login authentication for authorized users.

5. Main Modules / System Components

- **Login Module:** Handles user authentication.
- **Employee Module:** Handles adding, viewing, updating, and deleting employee records.
- **Search Module:** Helps users find employee records quickly.
- **Department Module:** Maintains basic department information.
- **Database:** Stores employee and user information.
- **User Interface:** Provides a simple interface for interacting with the system.

6. Non-Functional Requirements

- **Usability:** The system should be simple and easy to use.
- **Performance:** Employee records should be retrieved quickly.
- **Security:** Only authorized users should be able to modify employee data.
- **Reliability:** The system should store employee information accurately.
- **Maintainability:** The system should have a modular structure so that it can be easily modified.

7. Software Engineering Approach

The system will be developed incrementally using separate modules. Requirements will be identified first, followed by design, implementation, testing, and integration. Unit testing and integration testing will be performed to verify the correctness of individual modules and the complete system.

8. Technologies

- Programming Language: Python / Java
- Database: MySQL / SQLite
- User Interface: HTML, CSS and JavaScript / Java GUI
- Version Control: Git

9. Conclusion

The Employee Management System will provide a simple and centralized way to manage employee records. It will reduce manual work, make employee information easier to access, and provide basic security and data management facilities. Its modular design will also make the system easier to develop, test, and maintain.